

SomatoShift: A Wearable Haptic Display for Somatomotor Reconfiguration via Modifying Acceleration of Body Movement

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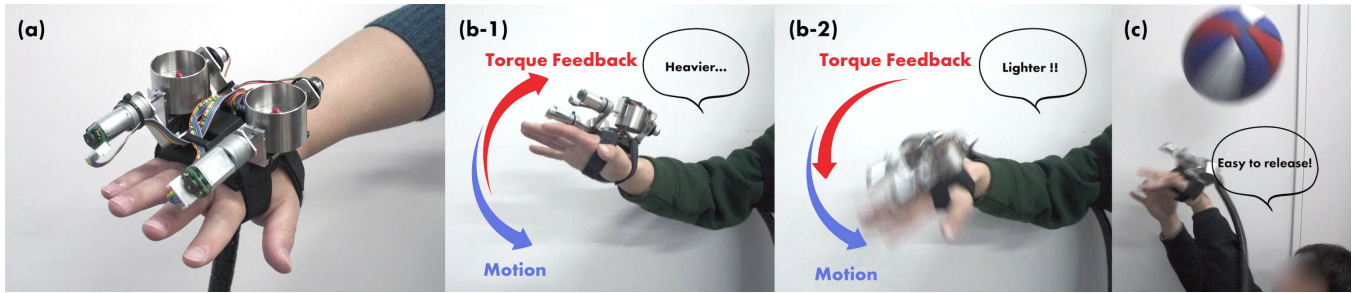


Figure 1: (a) SomatoShift is a wearable haptic display that utilizes two control moment gyroscopes. (b-1) When it applies a force against the user's motion, it gives an sensation of a heavier arm. (b-2) On the other hand, when the force aligns with the user's motion, it gives an sensation of a lighter arm. (c) This device has the potential not only to modify basic bodily sensations but also to transform our interaction with objects, including altering the manner in which one throws a basketball.

ABSTRACT

This paper proposes a wearable haptic device that utilizes control moment gyroscopes and a motion sensor to achieve *somatomotor reconfiguration*, altering the user's somatic perception of their body. The device can manipulate sensations, making body parts feel heavier or lighter, and modify the ease of movement during interactions with objects. Given its potential applications in avatar technology, sports, and assistive technology, this proposed device represents a promising avenue for enriching the user's bodily experiences.

KEYWORDS

Haptics; Proprioception; Body Perception; Embodied Interaction

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1 INTRODUCTION

Advancements in haptic technology have enabled the expression of haptic experiences in virtual reality, such as the sensation of touching an object or holding a tool. Haptic technologies have

found wide use in simulating interactions between humans and the physical world. Furthermore, ongoing research is exploring ways to allow individuals to experience bodily sensations other than their own [Nishida et al. 2020]. Virtual reality (VR) technology has facilitated the creation of avatars, visual representations that can embody multiple versions of oneself. By modifying not only an avatar's appearance but also its proprioceptive sensations — the internal bodily feelings — it becomes feasible to generate the impression of inhabiting a completely new body. The goal of this study is to modify participants' proprioceptive perception of their own bodies by appropriately intervening in their movements using a force feedback device. Our aim is to create an experience that goes beyond the physical limitations of the body, which is one of the many constraints humans. We believe this approach offers individuals an opportunity to liberate themselves from their physical bodily limitations.

2 SOMATOMOTOR RECONFIGURATION

Our system intervenes in a user's movement at strategic moments, giving rise to the sensation that the physical characteristics of their bodies have a change, as opposed to experiencing an external force being applied. We call this change in the relationship between somatosensory and motor sensations as *somatomotor reconfiguration*. To achieve this, we utilized impedance control, which modifies the static properties of an object as force information. Studies have shown that this can modify environmental properties [Hashimoto et al. 2022]. Similarly, we hypothesized that attaching a wearable force feedback device and providing force feedback in conjunction with movement could alter their joint motion, resulting in a change in body perception.

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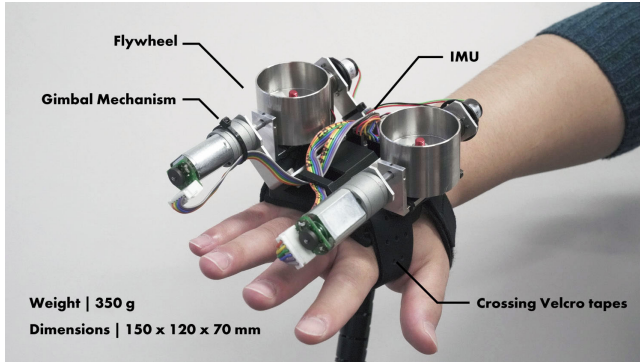


Figure 2: The device consists of two CMGs attached to a base component, and it can be worn on the body using a pair of intersecting Velcro straps.

3 SOMATOSHIFT

Essential components for realizing somatomotor reconfiguration consist of designing a wearable device and incorporating a Kinetic I/O system capable of human movement detection and immediate force feedback. In this study, we have fulfilled these requirements by developing a wearable device that uses an ungrounded force feedback technique.

3.1 Hardware

We have developed a wearable hardware prototype equipped with control moment gyroscopes (CMG), which produce torque from a spinning flywheel to deliver immediate force feedback in response to human movement. The mechanism and specifications of the device are shown in Figure 2. The weight of the wearable device was a critical consideration in its design, and following several prototype iterations, the final weight was brought down to 350 g.

Flywheel. The flywheel is fabricated from stainless steel to ensure high density, and it possesses a diameter of 40 mm, a mass of 30 g, and a moment of inertia of $1.27 \times 10^{-5} \text{ kgm}^2$. A brushless DC motor is employed to set the flywheel in motion, which consistently rotates at 10 000 rpm.

Gimbal Mechanism. We utilized a Pololu 63:1 geared DC motor and its encoder to drive the rotation of the gimbal mechanism. A slip ring was installed on the motor's opposite side to prevent cable entanglement.

Structure. The flywheel base and gimbal base are constructed from aluminum for strength and lightness, while the part worn on the hand is 3D-printed using PLA. The device can be securely attached to the hand by two intersecting Velcro tapes, which transmit the moments generated by the device to the hand.

IMU. The device is equipped with an Inertial Measurement Unit (IMU) to monitor body motion, and angular acceleration is calculated by differentiating the measured velocity from the IMU.

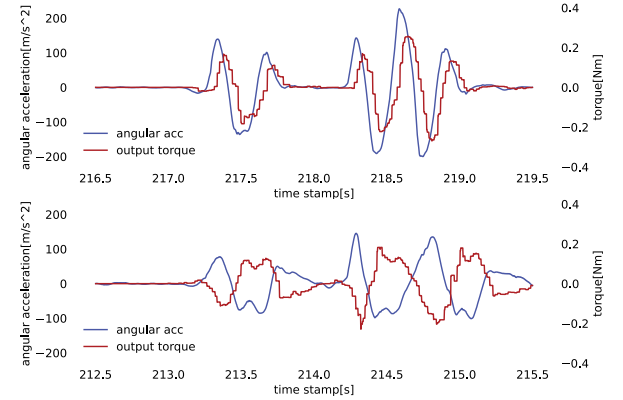


Figure 3: Angular acceleration (blue) and output torque (red) plot with upper graph showing $\Delta I < 0$ and lower graph showing $\Delta I > 0$.

3.2 Kinetic I/O system

The torque output τ in CMG systems is determined by the angle ϕ and angular velocity $\dot{\phi}$ of the gimbals. Our device uses a scissored paired CMG configuration, where two CMGs are positioned in parallel with each gimbal rotating in opposite phases. That is, the angle of each gimbal is represented as $(\phi_1, \phi_2) = (\phi, -\phi)$, and the torque output is $\tau = (0, 0, -2 \sin \phi)^T \dot{\phi}$. The velocity of the gimbal $\dot{\phi}$ to obtain the desired torque τ is calculated from the inverse problem of this equation. Following the concept of impedance control, our method outputs a torque proportional to the angular acceleration $\ddot{\omega}$ as $\tau = -\Delta I \ddot{\omega}$, where ΔI is a constant for how much motion is accelerated or decelerated.

4 TECHNICAL EVALUATION

Figure 3 displays the user's acceleration and estimated torque calculated from gimbal speed when utilizing the Kinetic I/O system. The torque can be observed to be outputted either in the same or opposite direction as the acceleration.

5 APPLICATION

This device holds potential for various applications, including integrating with visual VR avatars, enabling users to experience the bodily sensations of others, and facilitating learning of interactions with objects through one's own body.

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